

Exploring the Effects of Interactive Dialogue in Improving User Control for Explainable Online Symptom Checkers

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ABSTRACT

There has been a major push to improve the transparency of online symptom checkers (OSCs) by providing more explanations to users about their functioning and conclusions. However, not all users will want explanations about all aspects of these systems. A more user-centered approach is necessary for personalizing user experience of explanations. With this in mind, we designed and tested an interactive dialogue interface to afford user control to receive only those explanations that they would like to read. A user study ($N = 152$) with a text-based chatbot for assessing anxiety levels and presented explanations to participants in one of the three forms—an interactive dialogue providing choice for viewing different components of the explanations, a static disclosure of all explanations, and a control condition with no explanations whatsoever. We found that participants varied in the kinds of information they wanted to learn. The interactive delivery of explanations led to higher levels of perceived transparency and affective trust in the system. Furthermore, both subjective and objective understanding of the mechanism used for assessing anxiety was higher for participants in the interactive dialogue condition. We discuss theoretical and practical implications of imbuing interactivity for enhancing the effectiveness of explainable systems.

CCS CONCEPTS

• Human-centered computing → User studies.

KEYWORDS

online symptom checker, interactive dialogue, explanations, trust, user control

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1 INTRODUCTION

The growing demand for online healthcare and recent advances in artificial intelligence (AI) have fueled interest in automated health tools such as online symptom checkers (OSCs). These tools are available 24/7 for serving users at a time when they most need them, in the privacy of their homes [51]. For example, a large number of individuals have been using these OSCs to screen for and track mental health symptoms during the COVID-19 epidemic [12], lessening the burden on mental health practitioners [36]. However, the opacity of the diagnostic process of the OSCs has undermined user trust [17] and compliance to triage advice [7]. To address the “black-box” issues of such AI-powered health applications, considerable research attention has been devoted to exploring ways to improve transparency for medical AI in general [2, 35] and OSCs in particular [47, 51], with a focus on designing different types of explanations [47] or explanation techniques [35].

However, little effort has been paid to users’ preference for receiving explanations from these AI-enabled applications. Most current explanation methods follow a one-size-fits-all model [4], granting little user control over the learning process. Since users’ need for information may depend on individual differences and situational factors [28], different users may require different levels of details from the explanation. Due to the high-stakes nature of healthcare, users may seek control [13] and need to feel empowered in diagnosis processes [12] before they can understand the mechanisms and build trust in the OSCs [47, 50]. Therefore, giving users a more personalized learning experience of explanations is critical for building user understanding and trust [30]. Despite the rich literature in explainable recommender systems, such as ad suggestions [14], social media news feed curation [38], and music recommendation [29, 42], research on explainable AI in the context of health tools like OSCs is somewhat lacking. The resulting design challenge for explainable health systems is: How do we empower users in the process of explanation, in order to boost their understanding of and trust in them?

In response to this challenge, the current study aims to explore the effects of interactive dialogue for improving user control when processing explanations provided in the context of OSCs. An anxiety assessment questionnaire was administered through a rule-based health chatbot. The assessment mechanism was either revealed to the users through interactive dialogue, static interface, or without any explanation whatsoever. In the interactive dialogue condition, users could indicate whether they want to review the suggested scoring mechanism or not, and the chatbot provided follow-up information based on users’ previous choice. In this study, we investigate whether such interactive provision of information

enhances perceived transparency, user understanding and trust, as described in the sections that follow.

2 RELATED WORK

2.1 Transparency for Online Self Symptom Checkers

The opacity of AI-led health chatbots has resulted in “chatbot hesitancy” [32], which hinders acceptability of such applications. Through qualitative interviews, studies have discovered that users are concerned about the probing nature of the questions, which were “seemingly irrelevant” or “unreasonable” [51], raising concerns about “the sequence, quantity, and relevance of the questions” asked by the chatbot, thus confirming the dire need for explanations from these OSCs [47]. One study [47] examined different types of post-hoc explanations varied in rationale-based, feature-based and example-based explanations through an experiment, and found that all three explanations enhanced users’ perceived diagnostic quality and trust in diagnosis, but rationale-based explanation led to more cognitive overload that compromised user experience. Other studies have shown that interactive provision of information, as in a back-and-forth chat, can enhance user engagement [45]. It is therefore possible that introducing interactive dialogues into the supply of those explanations can be more effective in enhancing transparency.

Despite the increasing attention to the transparency issue, several questions remain: First, the current research on explanations from OSCs applies to a single message, which is presented all at once, even when it is interwoven into conversation flow (e.g., [7]). Such information, especially in detail, may create a feeling of “wall of text.” Being cognitively overloaded, users may easily neglect or mute such information, resulting in lower overall understanding of the AI system [1, 46], which can reduce their ability to form appropriate trust or make informed decisions. Second, the current explanations have followed a one-size-fits-all model [4], whereby the users have no control over what explanations they see. As Miller [28] argued, providing the same algorithmic explanation is not necessarily the best explanation. Users may hold a need for explanations with different levels of details, which can hinge upon individual differences such as cognitive style [28], or contextual factors such as issue relevance and urgency level [31]. Therefore, it has been suggested that users should be able to choose whether and what explanations they wish to see, and explanation components should be flexible enough to present varying level of details depending on a user’s preferences [16].

2.2 Achieving User Control through Interactive Dialogue

One way to address users’ need for control is through interactive dialogue, that is, the recipient requests the explanation through interactive communications. This also reinforces the view of Hilton [20] that “causal explanation takes the form of conversation and is thus subject to the rules of explanation.” In social science research, explanation is by nature a social interaction for transferring of knowledge through two-way communications [39]. Also, the power of dialogue is rooted in contingent conversation, where the system

is responsive to users when exchanging messages [45]. The back-and-forth interactions themselves have been found to enhance perceived transparency [49]. Empowering users with *control and freedom* has also been identified as one of the usability heuristics of conversational agents [22]. In our study, we examined whether providing explanations in user-driven interactive dialogue can have similar effects on explainable OSCs.

2.3 The Outcomes of Explanations: Objective Understanding, Subjective Understanding, and Trust

A critical question in transparency and explainable AI (XAI) research is how to make explanations understandable to a human [3]. While it is possible that providing information all at once may serve as a mental shortcut [46] that triggers the “illusion” of perceived understanding of the system, and can be powerful in building trust in AI systems [9, 10], it may not increase objective understanding, which is necessary for empowering users to make informed decisions. Interactive dialogue may hold the potential to achieve such objective understanding. Prior research has found that through conversations, individuals pay more attention to the messages and be motivated to process information in a systematic way [33]. Moreover, studies in cognitive learning have shown that when individuals have control over what information they can review, they tend to improve their knowledge more effectively [48].

Against this backdrop of the purpose of explanations, human explanatory preferences, and explanations in OSCs, this exploratory study seeks to answer the following research questions:

RQ1: Does explaining through interactive dialogue affect users’ perceptions of and trust in an OSC?

RQ2: How will users’ understanding of an OSC be affected by the explanation through interactive dialogue?

3 METHOD

3.1 Chatbot Prototype and Procedure

To test the effects of interactive dialogue in the context of explainable OSCs, we designed three versions of a chatbot prototype using the platform Landbot (<https://landbot.io/>). We provided explanations through either interactive dialogues or static interface, and contrasted both with a condition in which no explanations were offered to users. Participants were informed that they would interact with an AI-powered health chatbot for assessing anxiety levels. The chatbot applied the one-dimensional instrument of Generalized Anxiety Disorder-7 (GAD-7) designed to detect generalized anxiety disorder, which has been proven to be a valid and efficient tool for screening in clinical practice and research [25, 40, 43]. Upon obtaining consent, the chatbot started to ask GAD-7 questions, followed by two optional questions regarding gender and age. After receiving anxiety assessment and general advice on how to deal with their level of anxiety, participants were randomly assigned to one of three conditions: learning about the assessment mechanism through interactive dialogue, static interface, or no such information. The three conditions varied in the manner and extent to which the chatbot revealed how it arrived at a particular anxiety assessment score for the user (Fig.1).

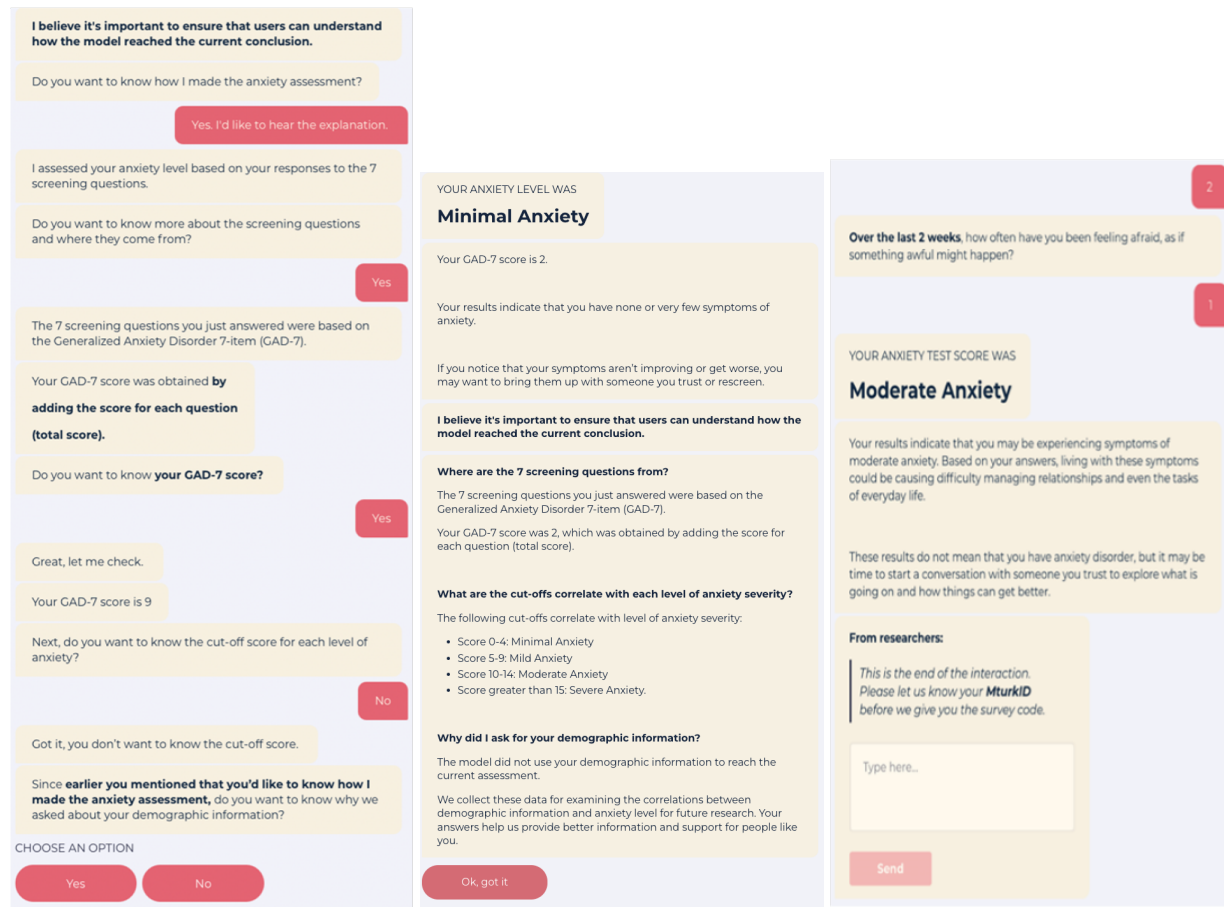


Figure 1: Illustration of providing explanations through interactive dialogue (left), static explanation condition (center), and no explanation condition (right).

In the interactive dialogue condition, the scoring mechanisms were divided into four different components pertaining to the anxiety assessment made by the chatbot: (1) where the 7 screening questions are from, (2) the user's GAD-7 score, (3) the cut-offs for each level of anxiety severity, and (4) why the chatbot asked for demographic information. The chatbot then asked several questions starting with "Do you want to know how I made the anxiety assessment?", and users could choose to read the offered piece of information or not. The chatbot ended the conversation if participants indicated "No" for the first question. Participants who chose "Yes" could keep learning other information. If the user chose to skip one particular piece of information, the chatbot nudged users to explore other information by referring to the user's previous inputs ("Previously, you mentioned ...") (Fig.1 left). In the static explanation condition, the chatbot displayed all the scoring algorithms at once (Fig.1 center). In the control condition, information regarding the scoring mechanism was absent (Fig.1 right). At the end of the session, the chatbot provided participants with a survey code, and redirected them to Qualtrics to complete the questionnaire. Participants were debriefed in this way: "Your interaction with the chatbot is not a formal diagnosis of your anxiety level. Please consult a trained professional, such as a doctor or a mental health provider, for anxiety determination."

3.2 Participants

The study was launched on CloudResearch upon receiving Institutional Review Board (IRB) approval. A total of 176 Mturkers participated in our study to receive monetary compensation. After excluding participants who did not complete the questionnaire ($n = 6$) and failed the attention check question ($n = 21$), the final sample consisted of 152 participants, with 87 females and 63 males, 2 declined to disclose, 62% of the participants self-identified as Caucasian/White, and 43% reported having a bachelor's degree or above. The average age was 37.39 ($SD = 10.3$).

3.3 Measures

All scales were measured with 7-point Likert scales unless otherwise indicated. Two aspects were measured for users' perceptions: *Perceived transparency* was assessed with four items adapted from [41, 47] including "The healthcare chatbot 'clearly explains how the chatbot makes the anxiety assessment' and 'shares relevant information about its algorithms to users'." *Perceived informativeness* was assessed by a semantic differential scale adopted from [11], measuring the extent to which the participants perceived the explanations as "helpful," "important," "informative," and "useful." In addition to the perceptions of the chatbot, we measured trust in

three dimensions: *affective trust* was assessed via 3 items adapted from [21] (e.g., “The chatbot displayed a warm and caring attitude toward me”); *cognitive trust* was measured with 3 items adapted from [34] (e.g., “This chatbot is reliable for assessing health-related issues”); and *behavioral trust* was assessed via 3 items adapted from [5] (e.g., “I intend to continue using this chatbot in the future”).

Perceived understanding was measured with 4 items adapted from [10, 37] (e.g., “I understood how the algorithms evaluated my anxiety level”). For *Objective understanding* of the system, we created five quiz questions to measure user understanding of the individual attributes on the output and how well they can predict the algorithm’s decision, based on previous research by [9], including dimensions such as recall information (e.g., “According to the chatbot, why did it ask you about your gender and age?”) and prediction of the model (e.g., “According to the cut-off score with each level of anxiety severity, if a user got a score of 8 in the screening questions, what would be his/her anxiety level?”). We summed up each participant’s total correct answers, ranging from 0 (all questions are wrong/I don’t know) to 5 (all questions are correct).

We also measured *need for cognition* (NFC) with 6 items such as “I would prefer complex to simple problems” from [23], since NFC may affect the propensity to scrutinize the information [28]. Lastly, we measured participants’ *familiarity with OSCs* (e.g., “I have experience interacting with OSCs”). The construct validity and reliability of all measures satisfied the criteria proposed by [15]. The average variance extracted (AVE) was higher than .50, and the composite reliability was higher than .70. Also, the AVE for each construct was greater than the correlation coefficients involving that construct, establishing good discriminant validity, as per a confirmatory factor analysis (CFA).

4 RESULTS

4.1 Chatbot Interaction Overview

To get an overview of the chatbot interactions, we linked each participant’s anxiety assessment result and chat history and linked them back to their assigned condition (interactive dialogue condition: $n = 50$; static condition: $n = 49$, control condition: $n = 52$). The results showed that 29% of the total participants had a minimal level of anxiety, 24% mild anxiety, 24% moderate anxiety, and 22% had severe anxiety. An analysis of variance (ANOVA) showed that the anxiety level did not differ across the three conditions, $F(2, 148) = 0.11, p = .89$.

In addition, within the interactive dialogue condition, 72% of them expressed interest in hearing how their anxiety level was assessed. A further analysis with logistic regression did not find significant effects of participants’ assessed anxiety level ($\chi^2 = .04, p = .85$) or NFC ($\chi^2 = .18, p = .67$) on their intentions to learn about the explanations. The one piece of information they most wanted to learn about was their exact assessment score, whereas the information about the screening questions was the least interesting to them (Table 1).

4.2 Effects of Explanation Style on User Perceptions and Trust (RQ1)

A series of analyses of covariance (ANCOVA) showed significant main effects of explanation style on perceived transparency, with

NFC and total interaction time as covariates, $F(2, 146) = 28.24, p < .001, \eta^2 = .28$. A post hoc analysis revealed that the interactive dialogue condition led to the highest degree of perceived transparency ($M = 5.71, SE = 0.17$), followed by the static interface ($M = 5.03, SE = 0.16$), with the lowest being the control condition ($M = 3.98, SE = 0.16$). A significant main effect was found for perceived informativeness, $F(2, 146) = 3.39, p = .04, \eta^2 = .044$. The post-hoc analyses showed that participants from the interactive dialogue condition ($M = 5.68, SE = 0.18$) and static condition ($M = 5.24, SE = 0.18$) perceived a significantly higher level of informativeness than the control condition ($M = 5.03, SE = 0.18$), though the difference between the interactive and static conditions was not significant. In addition, the main effect of the explanation style on affective trust was significant, $F(2, 146) = 3.2, p = .04, \eta^2 = .03$. Interactive dialogue condition contributed to the highest affective trust in the chatbot ($M = 4.78, SE = 0.22$) than static explanation ($M = 4.21, SE = 0.22$) and control condition ($M = 4.05, SE = 0.21$). However, the main effect was not significant for cognitive trust, $F(2, 146) = 1.75, p = .18$ or behavioral trust, $F(2, 146) = 0.93, p = .39$.

4.3 Effects of Explanation Style on Understandability (RQ2)

Our analysis also showed that users differed in terms of perceived understanding, $F(2, 146) = 22.08, p < .001, \eta^2 = 0.23$. Specifically, participants receiving explanations through interactive dialogues reported significantly higher perceived understandability ($M = 5.61, SE = 0.16$) than both static interface ($M = 4.94, SE = 0.17$) and control conditions ($M = 4.12, SE = 0.16$). The main effect was also significant for objective understanding of the system, $F(2, 146) = 20.59, p < .001, \eta^2 = 0.23$. Participants from the interactive dialogue condition ($M = 2.78, SE = 0.17$) answered a significantly higher number of quiz questions correctly compared to those who in the static explanation condition ($M = 1.84, SE = 0.18$) and those who did not receive any explanations whatsoever ($M = 1.22, SE = 0.17$). A simple linear regression showed that perceived understanding was a significant predictor of objective understanding ($b = 0.34, p < .001$).

4.4 Underlying Mechanisms for Enhancing Trust and Understandability

We conducted mediation analyses using PROCESS Macro (Model 4) [18] with perceived transparency, informativeness and understanding of the OSC as mediators. The indirect effects of interactive dialogue on cognitive trust ($b = .32, SE = 0.11, 95\%CI[.14, .52]$) and behavioral trust ($b = .40, SE = 0.13, 95\%CI[.17, .67]$) were significant, with perceived transparency as a positive mediator. Similarly, perceived informativeness was also found to be a positive mediator for cognitive trust ($b = .22, SE = 0.12, 95\%CI[.01, .49]$) and behavioral trust ($b = .35, SE = 0.19, 95\%CI[.01, .75]$). Furthermore, we found that perceived transparency was the only significant mediator to explain the effects of interactive dialogue on objective understanding ($b = .20, SE = 0.08, 95\%CI[.07, .37]$).

Table 1: Number of Participants seeking each of the Explanations in the Interactive Dialogue condition

Explanations in the Interactive Dialogue condition	N	Percentage
Do you want to know your GAD-7 score?	40	80%
Do you want to know how I made the anxiety assessment?	36	72%
Do you want to know why we asked about your demographic information?	27	54%
Do you want to know the cut-off score for each level of anxiety?	25	50%
Do you want to know more about the screening questions and where they come from?	19	38%

4.5 Examining the Effects of User Choice

To further examine the effects of user choice, we compared participants who chose to receive all the information from interactive dialogue condition ($n = 14$) with the other two conditions. The data showed these 14 participants scored significantly higher than those from the static condition and control condition in terms of perceived transparency, $F(2, 110) = 16.64, p < .001$, informativeness, $F(2, 110) = 4.31, p = .02$, subjective understanding, $F(2, 110) = 15.19, p < .001$, and objective understanding, $F(2, 110) = 34.47, p < .001$, but no significant results were found for trust. Additionally, participants who skipped all the information from the interactive dialogue condition ($n = 13$) performed similarly to those from the control condition for objective understanding, $F(1, 61) = 0.71, p = .40$, while they perceived significantly higher degree of understanding, $F(1, 61) = 5.92, p = .02$ and transparency $F(1, 61) = 11.06, p = .002$.

5 DISCUSSION

5.1 Summary of Findings and Theoretical Implications

First, the data showed that participants had different preferences regarding which detailed explanation they prefer to see. Specifically, our participants indicated the strongest interest in learning about their exact GAD-7 score, while they did not show much interest in learning about the source of the screening questions. These findings corroborated other studies [16, 30] which have claimed that users vary in the amount of information they need, based on their level of knowledge, skill, or curiosity.

Second, we found that providing explanations about the system by way of interactive dialogue can enhance positive user perceptions as well as trust in the OSCs, thus extending a previous study [47] which showed that mere presence of explanation can enhance perceived transparency and trust. Furthermore, our study sheds light on the underlying theoretical mechanism by showing that perceived transparency and informativeness mediate the effect of interactive dialogue upon user trust. By allowing users to choose among different pieces of information, users have more control over the explanations they receive, thus enhancing perceptions of transparency and informativeness, and thereby creating a sense of agency, which is a key driver of user experience in HCI research [26]. Similar effects have been found in the context of personalization [8]—merely asking users if they would like to receive a particular piece of information instead of providing it unsolicited can help promote trust. It also echoes the effects proposed by the action route of the Human-AI Interaction Model based on the Theory of Interactive Media Effects, i.e., the HAIL-TIME Model [44], which posits that deeper user interaction with an AI system can generate

more robust trust. Although we have known for some time that interactive explanations enhance transparency in recommender systems [19, 27], ours is the first attempt we know that applies interactive dialogue for providing diagnostic transparency in OSCs. Future research can examine how the relatively short explanations that are easy to communicate via interactive dialogue translate to contexts involving more complex models [50].

The third key result is that users' objective understanding of the anxiety assessment processes was higher in the interactive dialogue condition, even though many users opted to learn just the parts they were interested in and bypassed the rest. Furthermore, we found that both perceived understanding and objective understanding were enhanced by the interactive dialogue explanation style, in line with previous research [9]. Such effects may arise from the cognitive burden created by presenting all information regarding the explanations all at once [6]. It may also be possible that the one-page information can easily be ignored or skipped, and only served as a "transparency heuristic" [46]. Most importantly, our data showed that participants who received all information via interactive dialogue scored significantly higher than those from the static condition, which underscores the importance of user choice and control over the learning process. Interest is recognized as a crucial factor in cognitive learning [48]: When content is of high interest, learners are more motivated and therefore learn more [41].

5.2 Design Implications

The findings of our study have several design implications. First, as there was a significant drop-out rate for each prompt in the interactive dialogue design, it may be beneficial to survey users' need for information from OSCs when designing the information, to focus on those pieces of information that may be more helpful and seen as necessary by users. Furthermore, we found that users showed higher interest in learning their own diagnosis score than other general information, suggesting that more individualized information incorporating users' interaction history is more desirable, probably by imbuing a sense of uniqueness [24] compared to standard messaging. Second, our study demonstrated that when users had control over what information they can receive during the explanation process, they revealed better understanding of the scoring mechanism as well as more trust toward the OSC. Since we showed that interweaving the simple "Yes/No" choice option in the dialogues can significantly enhance user experience and trust, future work should investigate other possible designs for enhancing user control (without burdening them) when providing explanations. Designers of OSCs could consider implementing similar interfaces for users to indicate their preference. Third, designers could also leverage the power of dialogues from conversational agents as an ideal platform for providing explanations. OSCs can

engage users in conversations after giving assessment and medical advice. Web-based platforms that provide healthcare information and assessments can also embed chat agents to provide explanations or design interactive interfaces which allow users to ask for follow-up information.

5.3 Limitations

There are several limitations in this work. First, we employed an experimental study to draw causal conclusions, but it restricts our external validity. For future studies, we will investigate how users utilize the explanation interfaces in real-world settings. Second, our study was conducted during the pandemic with the topic of anxiety assessment, which may be more relevant given the surge in distress [86]. Further, it is possible that users' familiarity of GAD-7 led to the high drop-out rate in our study. Future studies should consider measuring knowledge of diagnosis mechanisms. Finally, the health chatbot we developed only applied standard screening methods with a simple rule-based algorithm to ensure experimental control. Future research can seek to replicate the results of the study with OSCs enabled by Natural Language Processing (NLP) or deep learning machine learning models for more complex diagnoses (e.g., [50]).

6 CONCLUSION

This study enhanced our understanding of the effects of interactive dialogues when processing explanations from OSCs. The study found that users hold different preferences for different types of explanations, especially the information that is individualized to them. In addition, our data indicated that by asking users to choose whether they want to receive a certain part of the scoring mechanism through interactive dialogue, they perceived a higher degree of transparency, and built more affective trust in the OSC. Lastly, we found that users who were only exposed to partial information from the interactive dialogue condition, or received the same amount of information with the static or no-explanation condition, still showed higher understanding of the OSC. The results of this preliminary user study demonstrate the potential benefits of employing interactive dialogue for enhancing the explainability and trustworthiness of OSCs in particular and AI systems in general.

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